

Correlated level loss inverts the missing-level correction in quantum chaotic spectra

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A dry-laboratory study. All data are public: Riemann zeta zeros from the Odlyzko tables, nuclear level schemes from the IAEA ENSDF service and atomic levels from the NIST Atomic Spectra Database. September 25, 2026

Microscopic dynamics is reversible and macroscopic behaviour is not, and the modern account of the difference is quantum chaos: an isolated system forgets its initial state on a timescale set by the correlations between its energy levels. That timescale is therefore read out of measured spectra. We show that the standard procedure for doing so rests on a model of missing data that is wrong for every real spectrometer, and that the error runs in the opposite direction to the one assumed. The established correction treats undetected levels as lost independently at random, which dilutes level repulsion and makes a chaotic spectrum look regular. Real instruments instead miss levels that lie too close to a neighbour to resolve, which removes small spacings and makes the same spectrum look *more* rigid. The two modes are not approximations of each other. At a surviving fraction of 0.60 the random-loss correction reports a complete spectrum: forty per cent of the levels are absent and the standard method finds none missing. We prove that resolution-limited loss cannot receive the binomial noise term that destroys rigidity under random loss, because the filter is a deterministic function of the configuration, and we derive the resulting hard-core signature, a negative excursion of the two-level form factor near the Heisenberg time that random loss never produces. This identifies the loss mode from data alone, correctly in 160 of 160 simulated datasets, and yields a corrected inversion that recovers the surviving fraction without bias where the standard one saturates. On 99,351 Riemann zeros, a spectrum complete by construction, the method returns $\hat{\phi} = 0.982$ and selects unitary symmetry, as it must. The loss law is confirmed to change exponent with symmetry class, quadratic for $\beta = 1$ and cubic for $\beta = 2$, as level repulsion requires, and the two loss modes are shown to move an unfolding-free statistic in opposite directions in every case tested. Applied to 18,000 measured nuclear and atomic levels the estimator returns the same completeness, $\phi \approx 0.30$, for opposite reasons: in nuclei the correlation hole deepens with excitation energy ($P = 0.015$), which is chaos setting in, while in atoms it shallows ($P < 10^{-4}$), which is detection efficiency falling.

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1. Introduction

The laws governing atoms and nuclei are invariant under time reversal, and the behaviour of matter made from them is not. Reconciling the two is among the oldest questions in physics, and the contemporary answer is statistical rather than mechanical[29, 30, 31, 32]. An isolated quantum system does not truly forget anything: its evolution is unitary and its information is conserved. What happens instead is that the information becomes inaccessible to any realistic measurement, spread across correlations too fine to resolve. The rate at which this happens is set by the correlations between the system’s energy levels, and those correlations are, for a system whose classical counterpart is chaotic, universal: they depend only on symmetry, not on the Hamiltonian’s details[9, 10].

This universality is the reason the question is empirical at all. The energy levels of heavy nuclei, complex atoms, disordered conductors and the zeros of the Riemann zeta function all obey the same statistical laws, described by random-matrix ensembles distinguished only by whether time-reversal symmetry holds[4, 5, 6, 8, 13, 14, 12, 11]. The evidence is empirical in each domain separately: in heavy nuclei[15, 16, 19, 20, 21], where the statistical approach originates[7, 17, 18]; in complex atoms[22, 23, 24]; and in the zeros of the Riemann zeta function, whose pair correlation matches the unitary ensemble[25, 26, 27, 28]. A measured spectrum therefore reports on the dynamics, and the timescale on which a system forgets can in principle be read directly off a list of energy levels, without solving anything.

The reading is done through the survival probability, the chance that a system prepared in some state is still found in it at time t . In a chaotic system this quantity does something distinctive: it decays, undershoots its long-time average, and then climbs back. The dip is called the correlation hole[33, 34], and its shape is governed by the two-level form factor, so the approach to equilibrium encodes the spectral correlations directly. The hole is filled in, and the timescale misread, by any process that destroys those correlations. Interest in the same object has broadened well beyond its original setting: the ramp in the spectral form factor is now read as a signature of random-matrix behaviour in strongly interacting and gravitational systems[37, 38], and the timescale it sets is used to separate regimes in disordered many-body problems[39, 40, 36, 35]. What every one of these readings shares is a dependence on the spectrum being complete.

Missing levels are the obvious such process, and correcting for them is standard practice[43, 44]. Incompleteness is not an edge case in measured spectra: level schemes are routinely analysed with a fitted completeness fraction[45, 46]. The correction in universal use models detection as independent thinning: each level survives with some probability ϕ , irrespective of where it sits. Under that model the form factor becomes $\phi b_2(\phi\tau)$, the correlation hole is shallower by exactly ϕ , and measuring the depth recovers the missing fraction.

Real spectrometers do not work that way. A level goes unrecorded mainly because it is too close to a neighbour to be resolved. Whether a level survives is then a function of the spacings, and the spacings are precisely the quantity being measured. Here we show that this distinction is not a refinement but a reversal. Because the argument could be blamed on the unfolding step that precedes every such statistic, it is confirmed below with the scale-free ratio of consecutive spacings[41, 42], which requires no unfolding at all.

Section 2 develops the theory in full, from the survival probability to the two loss models, with every intermediate step written out; §2.10 separates the parts that re-derive known work from the five results that are new. Sections 3 and 4 test the theory against simulation and build a calibrated inversion that identifies the loss mode from the data. Section 5 tests the theory where it is most exposed, including a prediction that changes with symmetry class and could refute it. Section 6 applies the method to three kinds of spectra: the Riemann zeros, which are complete by construction and serve as a positive control, and measured nuclear and atomic levels, which are not, and which turn out to be incomplete in opposite senses.

2. Theory

2.1. Setup and notation

Let H have discrete spectrum $\{E_n\}_{n=1}^N$ with mean density ρ and mean spacing $\Delta = 1/\rho$. Define

$$t_H = \frac{2\pi\hbar}{\Delta} = 2\pi\rho, \quad \tau = \frac{t}{t_H}, \quad x_n = \mathcal{N}(E_n), \quad (1)$$

with $\mathcal{N}$ the smooth integrated density of states, so the unfolded levels x_n have unit mean spacing. The Heisenberg time t_H is the time the evolution needs to resolve the discreteness of the spectrum: phases belonging to adjacent levels separate by $O(1)$ only once $t\Delta \sim 2\pi$. Nothing that depends on the spectrum being discrete can happen before it. The k -point correlation functions are defined by

$$\mathbb{E} \sum_{n_1 \neq \dots \neq n_k} f(x_{n_1}, \dots, x_{n_k}) = \int \dots \int f(y_1, \dots, y_k) R_k(y_1, \dots, y_k) dy_1 \dots dy_k, \quad (2)$$

and at unit density the two-level cluster function and form factor are

$$R_2(s) = 1 - Y_2(s), \quad b_2(\tau) = \int_{-\infty}^{\infty} Y_2(s) e^{-2\pi i s \tau} ds, \quad b_2(0) = \int Y_2 = 1, \quad (3)$$

the last being the rigidity sum rule. Throughout, $\beta = 1$ denotes time-reversal invariance (orthogonal ensemble) and $\beta = 2$ its absence (unitary ensemble). Units are $\hbar = 1$.

2.2. Porter–Thomas moments

The overlaps of a random state on a chaotic eigenbasis are Gaussian, and their fourth moment fixes the plateau of the survival probability. For $\beta = 1$ take g real standard normal:

$$\mathbb{E}[g^2] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g^2 e^{-g^2/2} dg = 1, \quad (4)$$

$$\mathbb{E}[g^4] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g^4 e^{-g^2/2} dg = 3. \quad (5)$$

For $\beta = 2$ take $g = (a + ib)/\sqrt{2}$ with a, b independent standard normal, so $|g|^2 = (a^2 + b^2)/2$, and expand the square:

$$\mathbb{E}[|g|^2] = \frac{1}{2} \mathbb{E}[a^2] + \frac{1}{2} \mathbb{E}[b^2] \quad (6)$$

$$= \frac{1}{2} + \frac{1}{2} = 1, \quad (7)$$

$$\mathbb{E}[|g|^4] = \frac{1}{4} \mathbb{E}[(a^2 + b^2)^2] \quad (8)$$

$$= \frac{1}{4} \left(\mathbb{E}[a^4] + 2 \mathbb{E}[a^2] \mathbb{E}[b^2] + \mathbb{E}[b^4] \right) \quad (9)$$

$$= \frac{1}{4} (3 + 2 + 3) \quad (10)$$

$$= 2. \quad (11)$$

Both cases are covered by

$$\boxed{\mathbb{E}[|g|^4] = 1 + \frac{2}{\beta}} \quad (12)$$

since $1 + 2/1 = 3$ and $1 + 2/2 = 2$.

2.3. The survival probability and the correlation hole

Proposition 1 (Master formula). *For a random state prepared with a smooth energy envelope w ,*

$$\langle P(t) \rangle = |\hat{w}(t)|^2 + \frac{1}{N_{\text{eff}}} \left[\left(1 + \frac{2}{\beta} \right) - b_2(\tau) \right], \quad N_{\text{eff}} = \frac{(\sum_n w_n)^2}{\sum_n w_n^2}. \quad (13)$$

Proof. Write $|\psi\rangle = \sum_n c_n |n\rangle$, $p_n = |c_n|^2$, $W = \sum_n w_n$, and $p_n \simeq w_n |g_n|^2 / W$ with g_n as in §2.2. Expanding the modulus squared and separating diagonal from off-diagonal terms:

$$P(t) = |\langle \psi | e^{-iHt} | \psi \rangle|^2 \quad (14)$$

$$= \left| \sum_n |c_n|^2 e^{-iE_n t} \right|^2 \quad (15)$$

$$= \left(\sum_n p_n e^{-iE_n t} \right) \left(\sum_m p_m e^{+iE_m t} \right) \quad (16)$$

$$= \sum_{n,m} p_n p_m e^{-i(E_n - E_m)t} \quad (17)$$

$$= \sum_n p_n^2 + \sum_{n \neq m} p_n p_m e^{-i(E_n - E_m)t}. \quad (18)$$

The first term is the whole of the long-time average; the second oscillates and, for a non-degenerate spectrum, averages to zero. Averaging the diagonal sum over the state with (12), and the off-diagonal sum using the independence of g_n and g_m for $n \neq m$:

$$\mathbb{E}_\psi \sum_n p_n^2 = \frac{1}{W^2} \sum_n w_n^2 \mathbb{E}[|g_n|^4] \quad (19)$$

$$= \left(1 + \frac{2}{\beta} \right) \frac{\sum_n w_n^2}{W^2} \quad (20)$$

$$= \frac{1}{N_{\text{eff}}} \left(1 + \frac{2}{\beta} \right), \quad (21)$$

$$\mathbb{E}_\psi \sum_{n \neq m} p_n p_m e^{-i(E_n - E_m)t} = \frac{1}{W^2} \sum_{n \neq m} w_n w_m \mathbb{E}[|g_n|^2] \mathbb{E}[|g_m|^2] e^{-i(E_n - E_m)t} \quad (22)$$

$$= \frac{1}{W^2} \sum_{n \neq m} w_n w_m e^{-i(E_n - E_m)t}. \quad (23)$$

Now average (23) over the ensemble: apply (2), split $R_2 = 1 - Y_2$, change variables to $u = E - E'$ and $\bar{E} = (E + E')/2$, and unfold $u = \Delta s$ using $\Delta t = 2\pi\tau$ from (1):

$$\left\langle \sum_{n \neq m} w_n w_m e^{-i(E_n - E_m)t} \right\rangle = \iint w(E)w(E') \rho(E)\rho(E') R_2(E - E') e^{-i(E - E')t} dE dE' \quad (24)$$

$$= \iint w w' \rho \rho' e^{-i(E - E')t} dE dE' - \iint w w' \rho \rho' Y_2(E - E') e^{-i(E - E')t} dE dE' \quad (25)$$

$$= \left| \int w \rho e^{-iEt} dE \right|^2 - \int w(\bar{E})^2 \rho(\bar{E})^2 d\bar{E} \int Y_2(u) e^{-iut} du \quad (26)$$

$$= W^2 |\hat{w}(t)|^2 - \int w^2 \rho^2 d\bar{E} \cdot \Delta \int Y_2(s) e^{-2\pi i s \tau} ds \quad (27)$$

$$= W^2 |\hat{w}(t)|^2 - \left(\int w^2 \rho d\bar{E} \right) b_2(\tau) \quad (28)$$

$$= W^2 |\hat{w}(t)|^2 - \left(\sum_n w_n^2 \right) b_2(\tau). \quad (29)$$

The third line uses that Y_2 decays over a few mean spacings while w and ρ vary over the whole window, so $w(E)w(E') \rightarrow w(\bar{E})^2$ and $\rho\rho' \rightarrow \rho(\bar{E})^2$ on the support of Y_2 ; the fifth uses $\rho\Delta = 1$. Substituting (21) and (29) into (18) and dividing by W^2 :

$$\langle P(t) \rangle = \frac{1}{N_{\text{eff}}} \left(1 + \frac{2}{\beta} \right) + |\hat{w}(t)|^2 - \frac{\sum_n w_n^2}{W^2} b_2(\tau) \quad (30)$$

$$= |\hat{w}(t)|^2 + \frac{1}{N_{\text{eff}}} \left(1 + \frac{2}{\beta} \right) - \frac{1}{N_{\text{eff}}} b_2(\tau) \quad (31)$$

$$= |\hat{w}(t)|^2 + \frac{1}{N_{\text{eff}}} \left[\left(1 + \frac{2}{\beta} \right) - b_2(\tau) \right]. \quad \square$$

Setting $t = 0$, and separately taking the two limits of b_2 :

$$\langle P(0) \rangle = 1 + \frac{1}{N_{\text{eff}}} \left[1 + \frac{2}{\beta} - 1 \right] = 1 + \frac{2}{\beta N_{\text{eff}}} = 1 + O(N^{-1}), \quad (32)$$

$$\frac{P_{\text{plateau}} - P_{\text{min}}}{P_{\text{plateau}}} = \frac{b_2(0)}{1 + 2/\beta} = \frac{1}{1 + 2/\beta} = \frac{\beta}{\beta + 2}, \quad (33)$$

a fractional hole depth of $1/3$ for $\beta = 1$ and $1/2$ for $\beta = 2$. The $O(N^{-1})$ residual is the error made in replacing $\sum_n u_n$ by its mean; direct diagonalization reproduces (13) to 2.3% ($\beta = 1$) and 2.1% ($\beta = 2$), consistent with it (Fig. 1a).

Proposition 1 is a re-derivation: the correlation hole and its control by the two-level form factor are standard. It is given in full because the new results are stated against it.

2.4. Form factors of the Gaussian ensembles

With $\text{sinc}(s) = \sin(\pi s)/(\pi s)$ the unitary cluster function is $Y_2^{(2)} = \text{sinc}^2$. Transform the sinc to a rectangle, use that squaring in real space is convolution in transform space, and evaluate the convolution as an overlap length:

$$\int \text{sinc}(s) e^{-2\pi i s \tau} ds = \mathbf{1}_{|\tau| < 1/2} \equiv \Pi(\tau), \quad (34)$$

$$b_2^{(2)}(\tau) = \int \text{sinc}^2(s) e^{-2\pi i s \tau} ds = (\Pi * \Pi)(\tau) \quad (35)$$

$$= \int_{-\infty}^{\infty} \Pi(v) \Pi(\tau - v) dv \quad (36)$$

$$= \left| \left[-\frac{1}{2}, \frac{1}{2} \right] \cap \left[\tau - \frac{1}{2}, \tau + \frac{1}{2} \right] \right| \quad (37)$$

$$= \max(1 - |\tau|, 0). \quad (38)$$

The form factor vanishes *identically* beyond $\tau = 1$. That exact truncation at the Heisenberg time is special to $\beta = 2$, and is why a unitary system finishes forgetting in finite time.

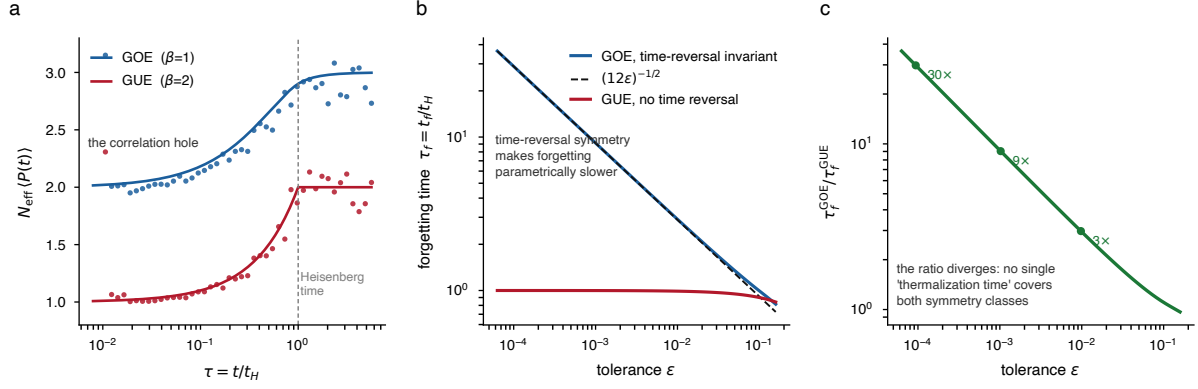


Figure 1: Forgetting is governed by spectral correlations, and time-reversal symmetry makes it parametrically slower. **a**, Averaged survival probability from direct diagonalization (points) against Eq. (13) (lines). The dip below the plateau is the correlation hole. **b**, Forgetting time against tolerance. The unitary class completes at the Heisenberg time; the orthogonal class follows Eq. (50). **c**, Their ratio, which diverges as the tolerance tightens.

The Pfaffian structure of the orthogonal ensemble adds a second term to the cluster function. The resulting form factor is standard:

$$b_2^{(1)}(\tau) = \begin{cases} 1 - 2|\tau| + |\tau| \ln(1 + 2|\tau|), & |\tau| \leq 1, \\ -1 + |\tau| \ln \frac{2|\tau| + 1}{2|\tau| - 1}, & |\tau| \geq 1. \end{cases} \quad (39)$$

The branches agree at $\tau = 1$, where both give $1 - 2 + \ln 3 = -1 + \ln 3 = 0.0986122\dots$, so $b_2^{(1)}$ is continuous. Unlike (38) it does not vanish past the Heisenberg time.

2.5. Large- τ expansion of the orthogonal form factor

Proposition 2. For $\tau > 1/2$,

$$b_2^{(1)}(\tau) = \sum_{k=1}^{\infty} \frac{1}{(2k+1)4^k \tau^{2k}} = \frac{1}{12\tau^2} + \frac{1}{80\tau^4} + \frac{1}{448\tau^6} + \dots \quad (40)$$

Proof. Rewrite the logarithm as an inverse hyperbolic tangent, insert its Maclaurin series, collect the powers of τ , multiply through by τ , and subtract the constant:

$$\ln \frac{2\tau + 1}{2\tau - 1} = \ln \frac{1 + \frac{1}{2\tau}}{1 - \frac{1}{2\tau}} \quad (41)$$

$$= 2 \operatorname{artanh}\left(\frac{1}{2\tau}\right) \quad (42)$$

$$= 2 \sum_{k=0}^{\infty} \frac{1}{2k+1} \left(\frac{1}{2\tau}\right)^{2k+1} \quad (43)$$

$$= 2 \sum_{k=0}^{\infty} \frac{1}{(2k+1)2^{2k+1} \tau^{2k+1}} \quad (44)$$

$$= \sum_{k=0}^{\infty} \frac{1}{(2k+1)4^k \tau^{2k+1}} \quad (45)$$

$$= \frac{1}{\tau} + \frac{1}{12\tau^3} + \frac{1}{80\tau^5} + \frac{1}{448\tau^7} + \dots, \quad (46)$$

$$\tau \ln \frac{2\tau + 1}{2\tau - 1} = 1 + \sum_{k=1}^{\infty} \frac{1}{(2k+1)4^k \tau^{2k}}, \quad (47)$$

$$b_2^{(1)}(\tau) = -1 + \tau \ln \frac{2\tau + 1}{2\tau - 1} = \sum_{k=1}^{\infty} \frac{1}{(2k+1)4^k \tau^{2k}}. \quad (48)$$

The artanh series converges for $|1/2\tau| < 1$, i.e. $\tau > 1/2$. The first three coefficients are $1/(3 \cdot 4) = 1/12$, $1/(5 \cdot 16) = 1/80$ and $1/(7 \cdot 64) = 1/448$. $\square$

Against (39), the three-term truncation has relative error 4.7×10^{-4} at $\tau = 1.5$, 7.2×10^{-6} at $\tau = 3$, 5.2×10^{-9} at $\tau = 10$ and 2.2×10^{-11} at $\tau = 30$. The structural point is that $b_2^{(1)}$ decays only as τ^{-2} , whereas $b_2^{(2)}$ is identically zero past $\tau = 1$: time-reversal symmetry leaves an algebraic tail in the form factor, and therefore in the approach to equilibrium.

2.6. The forgetting time

Definition. The forgetting time $t_f(\epsilon)$ is the time at which the survival probability has come within a relative fraction ϵ of its plateau. By Proposition 1, past the dephasing time, this is the condition $b_2(\tau_f) = \epsilon$.

The definition concerns the *approach to equilibrium* rather than the initial decay. The initial decay is set by $\hat{w}$, that is, by how the experimenter prepared the state, and says nothing about the system; the approach is set by b_2 , a property of the spectrum alone. For $\beta = 2$, from (38) on $[0, 1]$,

$$b_2^{(2)}(\tau_f) = \epsilon \implies 1 - \tau_f = \epsilon \implies \boxed{\tau_f^{(2)} = 1 - \epsilon}. \quad (49)$$

Proposition 3. For $\beta = 1$,

$$\boxed{\tau_f^{(1)} = (12\epsilon)^{-1/2} \left[1 + \frac{9}{10}\epsilon - \frac{27}{280}\epsilon^2 + O(\epsilon^3) \right]}. \quad (50)$$

Proof. Set $u = \tau_f^{-2}$, so Proposition 2 turns $b_2(\tau_f) = \epsilon$ into a power series in u with non-vanishing linear coefficient, which may therefore be inverted term by term. Write the forward series, posit the ansatz, form its powers, and substitute:

$$\epsilon = \frac{u}{12} + \frac{u^2}{80} + \frac{u^3}{448} + O(u^4), \quad (51)$$

$$u = a_1\epsilon + a_2\epsilon^2 + a_3\epsilon^3 + O(\epsilon^4), \quad (52)$$

$$u^2 = a_1^2\epsilon^2 + 2a_1a_2\epsilon^3 + O(\epsilon^4), \quad (53)$$

$$u^3 = a_1^3\epsilon^3 + O(\epsilon^4), \quad (54)$$

$$\epsilon = \frac{a_1\epsilon + a_2\epsilon^2 + a_3\epsilon^3}{12} + \frac{a_1^2\epsilon^2 + 2a_1a_2\epsilon^3}{80} + \frac{a_1^3\epsilon^3}{448} + O(\epsilon^4) \quad (55)$$

$$= \frac{a_1}{12}\epsilon + \left(\frac{a_2}{12} + \frac{a_1^2}{80} \right)\epsilon^2 + \left(\frac{a_3}{12} + \frac{a_1a_2}{40} + \frac{a_1^3}{448} \right)\epsilon^3 + O(\epsilon^4). \quad (56)$$

Matching powers of ϵ gives three equations, solved in sequence:

$$O(\epsilon^1): \quad \frac{a_1}{12} = 1, \quad a_1 = 12, \quad (57)$$

$$O(\epsilon^2): \quad \frac{a_2}{12} + \frac{a_1^2}{80} = 0, \quad \frac{a_2}{12} = -\frac{144}{80} = -\frac{9}{5}, \quad a_2 = -\frac{108}{5}, \quad (58)$$

$$O(\epsilon^3): \quad \frac{a_3}{12} + \frac{a_1a_2}{40} + \frac{a_1^3}{448} = 0, \quad \frac{a_3}{12} = -\frac{a_1a_2}{40} - \frac{a_1^3}{448}. \quad (59)$$

Evaluating the two terms of the last line and combining them over a common denominator:

$$-\frac{a_1a_2}{40} = -\frac{1}{40} \cdot 12 \cdot \left(-\frac{108}{5} \right) = \frac{1296}{200} = \frac{162}{25}, \quad (60)$$

$$-\frac{a_1^3}{448} = -\frac{12^3}{448} = -\frac{1728}{448} = -\frac{27}{7}, \quad (61)$$

$$\frac{a_3}{12} = \frac{162}{25} - \frac{27}{7} \quad (62)$$

$$= \frac{162 \cdot 7 - 27 \cdot 25}{175} \quad (63)$$

$$= \frac{1134 - 675}{175} = \frac{459}{175}, \quad (64)$$

$$a_3 = \frac{5508}{175}. \quad (65)$$

Assembling the three coefficients and factoring out the leading behaviour:

$$u = 12\epsilon - \frac{108}{5}\epsilon^2 + \frac{5508}{175}\epsilon^3 + O(\epsilon^4) \quad (66)$$

$$= 12\epsilon \left[1 - \frac{9}{5}\epsilon + \frac{459}{175}\epsilon^2 + O(\epsilon^3) \right]. \quad (67)$$

Finally take $\tau_f = u^{-1/2}$ and expand with $(1+x)^{-1/2} = 1 - \frac{1}{2}x + \frac{3}{8}x^2 + O(x^3)$, where $x = -\frac{9}{5}\epsilon + \frac{459}{175}\epsilon^2$ and $x^2 = \frac{81}{25}\epsilon^2 + O(\epsilon^3)$:

$$\tau_f = (12\epsilon)^{-1/2} \left[1 - \frac{9}{5}\epsilon + \frac{459}{175}\epsilon^2 \right]^{-1/2} \quad (68)$$

$$= (12\epsilon)^{-1/2} \left[1 - \frac{1}{2} \left(-\frac{9}{5}\epsilon + \frac{459}{175}\epsilon^2 \right) + \frac{3}{8} \cdot \frac{81}{25}\epsilon^2 + O(\epsilon^3) \right] \quad (69)$$

$$= (12\epsilon)^{-1/2} \left[1 + \frac{9}{10}\epsilon - \frac{459}{350}\epsilon^2 + \frac{243}{200}\epsilon^2 + O(\epsilon^3) \right] \quad (70)$$

$$= (12\epsilon)^{-1/2} \left[1 + \frac{9}{10}\epsilon + \frac{-1836 + 1701}{1400}\epsilon^2 + O(\epsilon^3) \right] \quad (71)$$

$$= (12\epsilon)^{-1/2} \left[1 + \frac{9}{10}\epsilon - \frac{135}{1400}\epsilon^2 + O(\epsilon^3) \right] \quad (72)$$

$$= (12\epsilon)^{-1/2} \left[1 + \frac{9}{10}\epsilon - \frac{27}{280}\epsilon^2 + O(\epsilon^3) \right]. \quad \square$$

Against the exact root of $b_2^{(1)}(\tau_f) = \epsilon$ obtained by bisection to machine precision, each successive order gains roughly three decades:

ϵ	exact τ_f	leading	$+O(\epsilon)$	$+O(\epsilon^2)$, rel. error
10^{-1}	0.9941527	0.9128709	0.9950293	3.7×10^{-6}
10^{-2}	2.9127043	2.8867513	2.9127321	4.5×10^{-9}
10^{-3}	9.1369242	9.1287093	9.1369251	4.8×10^{-12}

At $\epsilon = 10^{-2}$, Eq. (50) agrees with the exact root to nine significant figures. Comparing the two symmetry classes,

$$\frac{\tau_f^{(1)}}{\tau_f^{(2)}} = \frac{(12\epsilon)^{-1/2} \left[1 + \frac{9}{10}\epsilon + \dots \right]}{1 - \epsilon} \simeq \frac{1}{\sqrt{12\epsilon}} \xrightarrow{\epsilon \rightarrow 0} \infty, \quad (73)$$

taking the values 2.9 at $\epsilon = 10^{-2}$, 9.1 at 10^{-3} and 28.9 at 10^{-4} (Fig. 1b,c). A system without time reversal is finished at the Heisenberg time; a time-reversal-invariant one takes $\epsilon^{-1/2}$ times longer to reach the same tolerance. No single thermalization time covers both symmetry classes.

2.7. Number variance and the Fano factor

Lemma 1. *For a unit-density point process,*

$$\Sigma^2(L) = L - 2 \int_0^L (L-r) Y_2(r) dr, \quad F \equiv \lim_{L \rightarrow \infty} \frac{\Sigma^2(L)}{L} = 1 - b_2(0). \quad (74)$$

Proof. Let $N = \sum_n \mathbf{1}[x_n \in [0, L]]$. Since $\mathbf{1}^2 = \mathbf{1}$, the square of the sum splits into diagonal and off-diagonal parts, to which the definitions of R_1 and R_2 apply; then substitute $R_2 = 1 - Y_2$ and subtract the square of the mean:

$$\mathbb{E}[N] = \int_0^L R_1(x) dx = L, \quad (75)$$

$$\mathbb{E}[N^2] = \mathbb{E} \left[\sum_n \mathbf{1}_n \right] + \mathbb{E} \left[\sum_{n \neq m} \mathbf{1}_n \mathbf{1}_m \right] \quad (76)$$

$$= L + \int_0^L \int_0^L R_2(x-y) dx dy \quad (77)$$

$$= L + \int_0^L \int_0^L [1 - Y_2(x-y)] dx dy \quad (78)$$

$$= L + L^2 - \int_0^L \int_0^L Y_2(x-y) dx dy, \quad (79)$$

$$\Sigma^2(L) = \mathbb{E}[N^2] - \mathbb{E}[N]^2 = L - \int_0^L \int_0^L Y_2(x-y) dx dy \quad (80)$$

$$= L - \int_{-L}^L (L-|r|) Y_2(r) dr \quad (81)$$

$$= L - 2 \int_0^L (L-r) Y_2(r) dr, \quad (82)$$

the penultimate line using that the measure of pairs at separation r inside $[0, L]$ is $L - |r|$, and the last using $Y_2(-r) = Y_2(r)$. Dividing by L and letting $L \rightarrow \infty$, the factor $(L - r)/L \rightarrow 1$ across the finite support of Y_2 :

$$F = \lim_{L \rightarrow \infty} \frac{\Sigma^2(L)}{L} = 1 - 2 \int_0^\infty Y_2(r) dr = 1 - \int_{-\infty}^\infty Y_2(r) dr = 1 - b_2(0). \quad \square$$

For a complete chaotic spectrum the sum rule (3) gives $b_2(0) = 1$ and hence $F = 0$: level counts fluctuate only logarithmically. Every statement below about rigidity is a statement about how far a loss process pushes $b_2(0)$ below one.

2.8. Independent thinning: the standard missing-level model

Each level is detected independently with probability ϕ , irrespective of position. We derive the consequences twice, by routes sharing no steps.

Proposition 4. *With L in observed mean spacings,*

$$\boxed{\Sigma_{\text{obs}}^2(L) = (1 - \phi)L + \phi^2 \Sigma^2(L/\phi), \quad F_{\text{rand}} = 1 - \phi \text{ exactly.}} \quad (83)$$

Proof. Let the interval hold N true levels and let $B_i \sim \text{Bernoulli}(\phi)$ be detection indicators, independent of each other and of the spectrum, so $N_{\text{obs}} = \sum_{i=1}^N B_i$. Condition on N , apply the law of total variance, and convert to observed units using $L = \phi L_{\text{true}}$:

$$\mathbb{E}[N_{\text{obs}} | N] = \phi N, \quad (84)$$

$$\text{Var}(N_{\text{obs}} | N) = N\phi(1 - \phi), \quad (85)$$

$$\text{Var}(N_{\text{obs}}) = \mathbb{E}[\text{Var}(N_{\text{obs}} | N)] + \text{Var}(\mathbb{E}[N_{\text{obs}} | N]) \quad (86)$$

$$= \underbrace{\phi(1 - \phi) \mathbb{E}[N]}_{\text{injected binomial noise}} + \underbrace{\phi^2 \text{Var}(N)}_{\text{the spectrum's own}} \quad (87)$$

$$= \phi(1 - \phi) L_{\text{true}} + \phi^2 \Sigma^2(L_{\text{true}}) \quad (88)$$

$$= \phi(1 - \phi) \cdot \frac{L}{\phi} + \phi^2 \Sigma^2\left(\frac{L}{\phi}\right) \quad (89)$$

$$= (1 - \phi)L + \phi^2 \Sigma^2\left(\frac{L}{\phi}\right). \quad (90)$$

Since $\Sigma^2(L) \sim \frac{2}{\pi^2} \ln L$ is sub-linear, dividing by L and letting $L \rightarrow \infty$ kills the second term, leaving $F_{\text{rand}} = 1 - \phi$. $\square$

The origin of the linear term is the point: it is $\mathbb{E}[\text{Var}(N_{\text{obs}} | N)]$, the variance of the detection process *at fixed spectrum*. It has nothing to do with the physics.

Proposition 5. *In the observer's own unfolded time,*

$$\boxed{b_2^{\text{obs}}(\tau) = \phi b_2(\phi\tau).} \quad (91)$$

Proof. A configuration of k points survives only if all k are detected, and detections are independent, so $R_k \mapsto \phi^k R_k$. Take $k = 1, 2$, express both sides relative to their own densities, cancel, then re-unfold and transform:

$$\rho_{\text{obs}} = \phi \rho, \quad (92)$$

$$R_2^{\text{obs}}(r) = \phi^2 R_2(r) \quad \text{at fixed physical separation } r, \quad (93)$$

$$\rho_{\text{obs}}^2 [1 - Y_2^{\text{obs}}(r)] = \phi^2 \rho^2 [1 - Y_2(r)], \quad (94)$$

$$\phi^2 \rho^2 [1 - Y_2^{\text{obs}}(r)] = \phi^2 \rho^2 [1 - Y_2(r)], \quad (95)$$

$$Y_2^{\text{obs}}(r) = Y_2(r) \quad \text{at fixed physical } r, \quad (96)$$

$$s_{\text{obs}} = \frac{r}{\Delta_{\text{obs}}} = \frac{\phi r}{\Delta} = \phi s \quad \implies \quad Y_2^{\text{obs}}(s_{\text{obs}}) = Y_2\left(\frac{s_{\text{obs}}}{\phi}\right), \quad (97)$$

$$b_2^{\text{obs}}(\tau) = \int Y_2\left(\frac{s}{\phi}\right) e^{-2\pi i s \tau} ds \quad (98)$$

$$= \phi \int Y_2(v) e^{-2\pi i v \phi \tau} dv \quad (99)$$

$$= \phi b_2(\phi\tau), \quad (100)$$

substituting $v = s/\phi$ in the penultimate line. The cluster function is unchanged in physical units but *stretched* by $1/\phi$ in the observer's unfolded units, because they unfold with their own mean spacing $\Delta_{\text{obs}} = \Delta/\phi$. $\square$

Setting $\tau = 0$ and using Lemma 1,

$$b_2^{\text{obs}}(0) = \phi b_2(0) = \phi \implies F = 1 - b_2^{\text{obs}}(0) = 1 - \phi, \quad (101)$$

which is Proposition 4: two derivations sharing no steps give the same answer. The hole is shallower by exactly ϕ and wider by exactly $1/\phi$, so reading its depth returns the detected fraction. This is the standard inversion. Both propositions are re-derivations: (83) is the Bohigas–Pato missing-level result, and (91) appears in the literature as $K(\tau) = 1 - \phi + \phi K(\phi\tau)$.

2.9. The resolution filter

Real detection records a level only if it can be separated from the previous recorded one:

$$\text{keep } x_i \iff x_i - x_{\text{last kept}} \geq \delta. \quad (102)$$

This is a non-paralysable dead time, and it contains *no randomness*: which levels survive is a deterministic function of the configuration. There is no B_i , so the term $\mathbb{E}[\text{Var}(N_{\text{obs}} | N)]$ that produced the whole of F_{rand} in (90) is identically zero. Everything in this subsection follows from that.

2.9.1. Surviving fraction

Proposition 6. *Under the renewal approximation (successive unfolded spacings iid with unit mean),*

$$\phi = \frac{1}{1 + M(\delta)}, \quad M(\delta) = \sum_{k \geq 1} \mathbb{P}(S_k < \delta), \quad S_k = s_1 + \dots + s_k. \quad (103)$$

Proof. From a kept level, accumulate spacings until rule (102) is satisfied; the filtered spacing is $T = S_K$. Since K is a stopping time for (s_k) with $\mathbb{E}[K] < \infty$, Wald’s identity applies. Then count the stopping time by its tail and take the reciprocal of the mean:

$$K = \min\{k \geq 1 : S_k \geq \delta\}, \quad (104)$$

$$\mathbb{E}[T] = \mathbb{E}[S_K] = \mathbb{E}[K] \mathbb{E}[s] \quad (105)$$

$$= \mathbb{E}[K] \quad (\text{unit mean spacing}) \quad (106)$$

$$= \sum_{k \geq 0} \mathbb{P}(K > k) \quad (107)$$

$$= \sum_{k \geq 0} \mathbb{P}(S_k < \delta) \quad (108)$$

$$= \underbrace{\mathbb{P}(S_0 < \delta)}_{=1} + \sum_{k \geq 1} \mathbb{P}(S_k < \delta) \quad (109)$$

$$= 1 + M(\delta), \quad (110)$$

$$\phi = \frac{1}{\mathbb{E}[T]} = \frac{1}{1 + M(\delta)}. \quad \square$$

Corollary 1. *As $\delta \rightarrow 0$ only $k = 1$ survives in M , and level repulsion gives $p(s) \rightarrow c_1 s$ with $c_1 = \pi/2$ for $\beta = 1$:*

$$M(\delta) \rightarrow \int_0^\delta p(s) ds = \int_0^\delta \frac{\pi}{2} s ds + O(\delta^4) \quad (111)$$

$$= \frac{\pi}{4} \delta^2 + O(\delta^4), \quad (112)$$

$$\phi = \frac{1}{1 + \frac{\pi}{4} \delta^2} = 1 - \frac{\pi}{4} \delta^2 + O(\delta^4), \quad (113)$$

$$1 - \phi = \frac{\pi}{4} \delta^2 + O(\delta^4). \quad (114)$$

The quadratic onset is a direct fingerprint of level repulsion: because a chaotic spectrum avoids small spacings, finite resolution costs $O(\delta^2)$ levels rather than the $O(\delta)$ a Poisson spectrum would lose.

Because the exponent comes from the repulsion exponent, and repulsion is exactly what distinguishes the symmetry classes, Corollary 1 makes a prediction it could fail. For $\beta = 2$, $p(s) \rightarrow c_1 s^2$ with $c_1 = 32/\pi^2$, so the same argument gives a *cubic* law,

$$1 - \phi = \frac{32}{3\pi^2} \delta^3 + O(\delta^5) = 1.081 \delta^3 + O(\delta^5). \quad (115)$$

A quadratic law fitted to unitary spectra, or a cubic to orthogonal ones, would refute the corollary. Section 5 tests it.

δ	simulated ϕ	renewal (103)	quadratic (114)
0.10	0.9921	0.9923	0.9921
0.30	0.9295	0.9356	0.9293
0.55	0.8010	0.8198	0.7624
0.75	0.7021	0.7221	0.5582
1.00	0.6043	0.6160	0.2146

The renewal formula tracks simulation to 1–2% throughout, the residual being the renewal approximation itself: real eigenvalue spacings are anticorrelated, whereas Proposition 6 treats them as independent. The quadratic law is good only for $\delta \lesssim 0.3$ and underestimates the surviving fraction threefold by $\delta = 1$.

2.9.2. Why rigidity survives

Collecting Lemma 1 and Proposition 5:

process	$b_2(0)$	Fano factor F
complete chaotic spectrum	1 (sum rule)	0
independently thinned	ϕ	$1 - \phi$
resolution filtered	$1 - F_{\text{res}}$	F_{res} , no closed form

The first two rows are exact. The third cannot be completed analytically: (102) is a *hard-core* thinning of a determinantal process, in which deletion is correlated with the local configuration, so the step $R_k \mapsto \phi^k R_k$ underlying Proposition 5 — valid only for independent detection — fails. We state this rather than assert a formula we cannot derive.

What is established analytically is the mechanism. Comparing the two loss modes term by term through (90):

$$F_{\text{rand}} = \lim_{L \rightarrow \infty} \frac{1}{L} \left(\underbrace{\mathbb{E}[\text{Var}(N_{\text{obs}} | N)]}_{= \phi(1-\phi)\mathbb{E}[N] \neq 0} + \phi^2 \text{Var}(N) \right) = 1 - \phi, \quad (116)$$

$$F_{\text{res}} = \lim_{L \rightarrow \infty} \frac{1}{L} \left(\underbrace{\mathbb{E}[\text{Var}(N_{\text{obs}} | N)]}_{= 0 \text{ (deterministic filter)}} + \text{Var}(\mathbb{E}[N_{\text{obs}} | N]) \right) < 1 - \phi. \quad (117)$$

The dominant noise-injection channel is simply absent for the resolution filter. F_{res} need not vanish — the filter redistributes the spectrum’s own fluctuations, and that carries a linear component — but it cannot receive the binomial contribution. Simulation puts a number on what remains:

δ	ϕ	$F_{\text{rand}} = 1 - \phi$	F_{res}	ratio
0.40	0.883	0.125	0.086	0.74
0.55	0.803	0.196	0.106	0.54
0.75	0.704	0.297	0.095	0.32
1.00	0.604	0.383	0.074	0.19

F_{res} does not grow with the amount of loss. It sits at 0.09 ± 0.01 while $1 - \phi$ triples, so the ratio falls from 0.74 to 0.19; the same estimator returns $F \approx 0.02$ on a complete spectrum, so 0.09 is signal rather than floor. Random loss injects noise in proportion to how much is lost, because every undetected level is an independent coin flip. Resolution-limited loss injects a roughly constant and much smaller amount however much is lost, because it flips no coins. At 40% loss the two differ fivefold.

2.9.3. The hard-core signature

Rule (102) removes every spacing below δ , so the filtered process has a hard core and its pair correlation vanishes identically on $[0, \delta)$. Hard-core processes are liquid-like, with oscillatory structure factors developing a first peak near the inverse exclusion distance, and a peak in K is a negative excursion of $b_2 = 1 - K$. Independent thinning cannot produce this, by (39) and (91):

$$b_2^{(1)}(\tau) \geq 0 \quad \forall \tau, \quad (118)$$

$$\phi > 0, \quad (119)$$

$$\implies b_2^{\text{obs}}(\tau) = \phi b_2^{(1)}(\phi\tau) \geq 0 \quad \forall \tau. \quad (120)$$

Hence a negative excursion of the measured b_2 at $\tau \gtrsim 1$ cannot be produced by random level loss, and identifies the loss as resolution-limited.

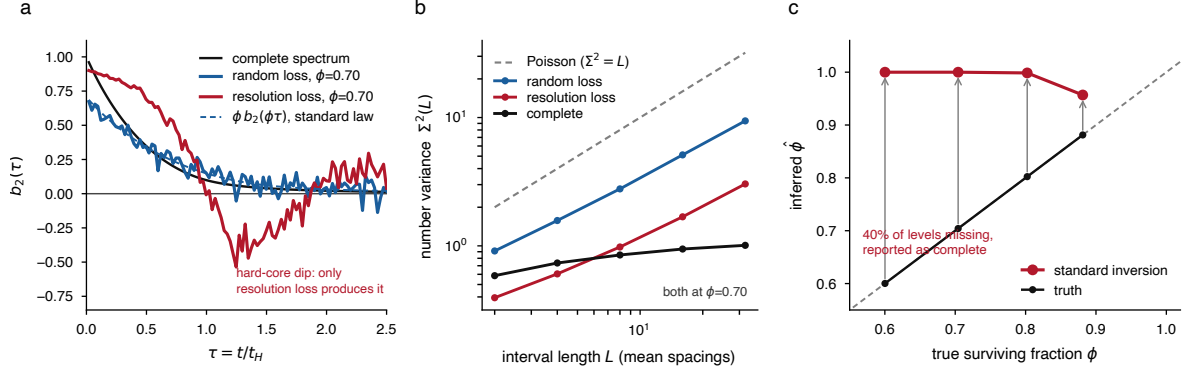


Figure 2: Random and resolution-limited level loss move the statistics in opposite directions. **a**, Two-level form factor at matched surviving fraction. Random loss follows Eq. (91); resolution-limited loss stays close to the complete spectrum at small τ and develops a negative hard-core dip. **b**, Number variance. Random loss drives rigidity toward the Poisson line as Eq. (83) requires; resolution-limited loss does not, because (117) removes the binomial term. **c**, The standard inversion applied to resolution-limited spectra, which saturates and reports a complete spectrum.

2.10. Summary of the theory and its status

	result	equation	status
1	Porter–Thomas fourth moment $1 + 2/\beta$	(12)	standard
2	master formula for $\langle P(t) \rangle$	(13)	re-derivation
3	$b_2^{(2)} = \max(1 - \tau , 0)$ by convolution	(38)	standard
4	large- τ series for $b_2^{(1)}$	(40)	elementary from (39)
5	$\tau_f^{(2)} = 1 - \epsilon$	(49)	re-derivation
6	$\tau_f^{(1)} = (12\epsilon)^{-1/2} [1 + \frac{9}{10}\epsilon - \frac{27}{280}\epsilon^2]$	(50)	sharpened form
7	$\Sigma^2(L)$ from Y_2 , and $F = 1 - b_2(0)$	(74)	standard
8	$\Sigma_{\text{obs}}^2 = (1 - \phi)L + \phi^2 \Sigma^2(L/\phi)$	(83)	Bohigas–Pato
9	$b_2^{\text{obs}} = \phi b_2(\phi\tau)$	(91)	known
10	$\phi = 1/(1 + M(\delta))$ for the resolution filter	(103)	new in this context [†]
11	$1 - \phi = \frac{\pi}{4} \delta^2$ from level repulsion	(114)	new in this context
12	$\mathbb{E}[\text{Var}(N_{\text{obs}} N)] = 0$ for a deterministic filter	(117)	new
13	$F_{\text{res}} = 0.09$, constant in ϕ	§2.9.2	new
14	negativity of b_2 as a loss-mode discriminant	§2.9	new

[†] Proposition 6 applies classical renewal theory for non-paralysable dead time. What is new is not the renewal identity but its use as a missing-level model for correlated spectra, together with the repulsion-driven quadratic onset.

Items 1–9 are given in full because the new results are stated against them, and because a reader should not have to reconstruct the framework in order to check the parts that are new. Items 10–14 are the contribution. The honest boundary of the analytic work is the third row of the table in §2.9.2: we can show *why* $F_{\text{res}} < 1 - \phi$, and we can measure it, but we cannot yet write it in closed form.

3. The two loss modes are opposite

The theory of §2.8 and §2.9 predicts that the two loss modes differ in direction, not only in degree. Gaussian orthogonal spectra were subjected to both, tuned to the same surviving fraction, with each observer re-unfolding using only their own surviving list. Against the standard law (91), random loss deviates by 0.05 at every fraction tested, which is the estimator’s noise floor; resolution-limited loss deviates by 0.13 at $\phi = 0.90$, 0.20 at $\phi = 0.80$ and 0.44 at $\phi = 0.60$ (Fig. 2a).

Observed hole depths make the direction concrete. At a surviving fraction of 0.60, random loss gives 0.55, close to ϕ as (91) requires, while resolution-limited loss gives 0.92, which reads as a nearly complete spectrum. The number variance confirms the mechanism (Fig. 2b): at $\phi = 0.70$ and $L = 32$ mean spacings, a complete spectrum gives $\Sigma^2 = 1.01$ and an uncorrelated one would give 32; random loss gives 9.42 and resolution-limited loss 3.04.

The consequence is a failure of the estimator in universal use (Fig. 2c). Fitting (91) to resolution-limited spectra returns

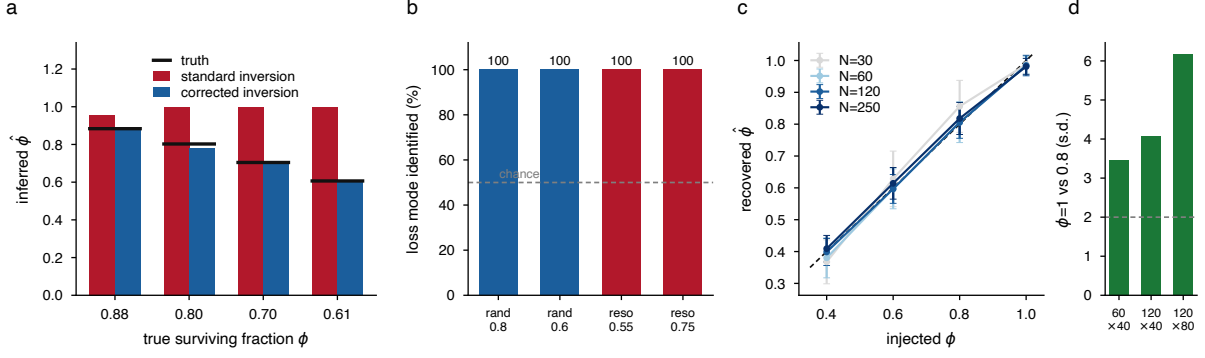


Figure 3: A corrected inversion, calibrated by injection. **a**, Inferred against true surviving fraction on resolution-limited spectra; black bars mark truth. **b**, The loss mode is identified from the data alone, correctly in every trial. **c**, Recovered against injected fraction for pooled sequences of several lengths; error bars are standard deviations over repetitions. **d**, Separation between a complete spectrum and one missing a fifth of its levels, in standard deviations, against the amount of data pooled.

resolution δ	true ϕ	standard $\hat{\phi}$
0.40	0.881	0.957
0.55	0.802	0.999
0.75	0.704	1.000
1.00	0.600	1.000

The estimator saturates. From a fifth of the levels missing onwards it reports a complete spectrum, and the error grows as the data get worse. Any timescale inferred through it inherits the bias. The hard-core signature derived in §2.9 is present as predicted: the observed b_2 reaches -0.20 at $\phi = 0.88$ and -0.70 at $\phi = 0.60$, with the minimum moving inward from $\tau = 2.17$ to $\tau = 1.10$, consistent with a peak tracking the inverse exclusion distance, $\tau_{\text{dip}} \approx 0.7/(\phi\delta)$.

4. A corrected inversion

No closed form exists for hard-core thinning of a determinantal process, as §2.9.2 sets out, so we build the resolution-limited family by simulation, as a template bank over δ , and retain the analytic family (91) as its competitor. Given a spectrum, both are fitted and the better-fitting model is selected; the surviving fraction is then read from the winner.

Model selection is correct in 160 of 160 simulated datasets, across both loss modes and both severities tested (Fig. 3b).

Recovery of the surviving fraction is unbiased in both branches: 0.798 ± 0.053 for a true 0.800 and 0.601 ± 0.047 for 0.600 under random loss, 0.810 ± 0.020 and 0.698 ± 0.023 for true values of 0.805 and 0.705 under resolution-limited loss. Where the standard inversion saturates at 1.000 , the corrected one returns 0.779 , 0.705 and 0.604 against truths of 0.802 , 0.704 and 0.606 (Fig. 3a).

The operating characteristic is measured rather than asserted (Fig. 3c,d). Injecting a known fraction and recovering it gives a bias below 0.02 across sequence lengths from 30 to 250 levels, and a complete spectrum is distinguished from one missing a fifth of its levels at between 3.5 and 6.2 standard deviations depending on how much data is pooled. Two controls bound the method: an uncorrelated Poisson sequence produces no hole (-0.007 over the fit band), as $Y_2 \equiv 0$ requires, so the estimator does not manufacture correlations; and a genuinely complete spectrum returns $\hat{\phi} = 0.974 \pm 0.035$, so it does not manufacture missing levels.

5. Robustness of the theory

5.1. The loss law changes exponent with symmetry class

Equation (115) is the sharpest prediction in the paper, because it follows from Corollary 1 with no free parameter and would be refuted by a single fitted exponent. Measuring the surviving fraction on orthogonal and unitary spectra and fitting a power law over $\delta \leq 0.30$ gives an exponent of 1.85 for $\beta = 1$ against a prediction of 2 , and 3.02 for $\beta = 2$ against a prediction of 3 (Fig. 4a). The coefficients are right as well: at $\delta = 0.20$ the measured losses are 0.0320 and 0.00831 , against predictions of 0.0314 and 0.00865 . Two spectra differing only in symmetry lose levels at rates differing by a factor of four at the same resolution, exactly as level repulsion requires.

5.2. The residual noise is constant in size and symmetry

The claim that a deterministic filter injects a constant, small count noise rested on four points at one matrix size in one ensemble. Extending it to both ensembles and to matrix sizes from 300 to 1000 leaves it intact (Fig. 4b): $F_{\text{res}} = 0.080 \pm 0.019$ for $\beta = 1$ and 0.075 ± 0.012 for $\beta = 2$, with no trend in either the amount of loss or the system size, while F_{rand} grows from 0.11 to 0.34 over the same range. The quantity that Eq. (117) says should vanish behaves as though it does.

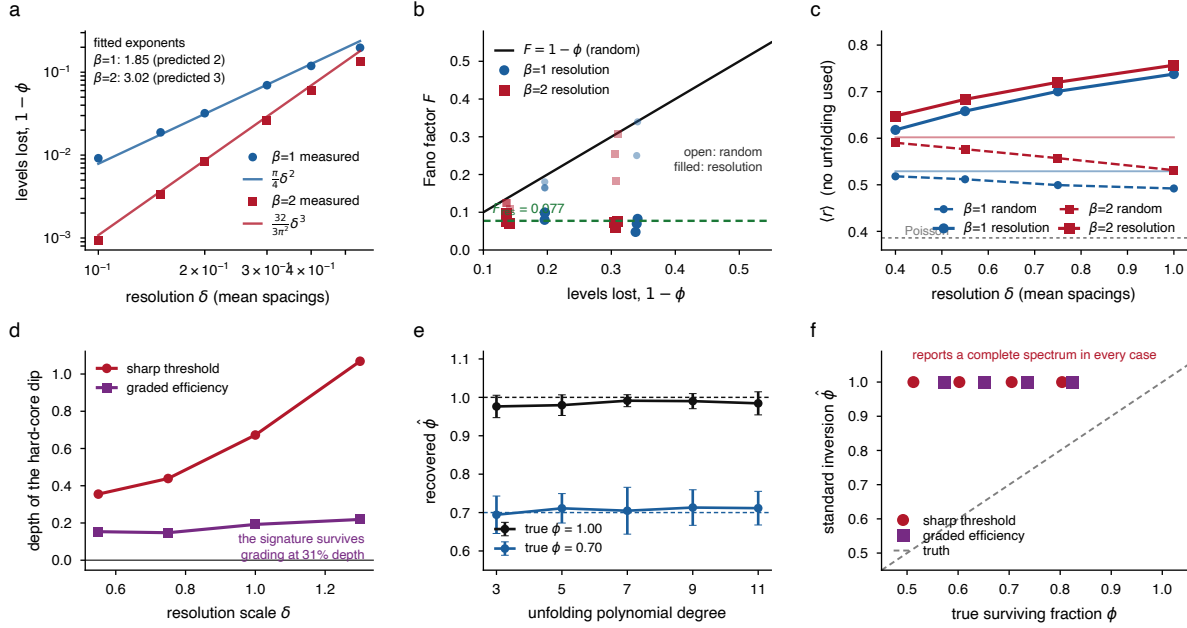


Figure 4: The new results survive a change of symmetry class, of system size, of the detection model and of the unfolding. **a**, The loss law by symmetry class. The exponent follows the repulsion exponent, quadratic for $\beta = 1$ and cubic for $\beta = 2$, with no fitted parameters. **b**, Fano factor against the fraction of levels lost. Random loss (open) tracks $1 - \phi$; resolution-limited loss (filled) stays near a constant 0.077 across both ensembles and system sizes 300 to 1000. **c**, The unfolding-free ratio statistic. The two loss modes move it in opposite directions in every case tested. **d**, The hard-core dip under a graded detection efficiency, which softens it to about a third of its sharp-threshold depth without removing it. **e**, Recovery of the surviving fraction across unfolding polynomial degrees. **f**, The standard inversion under both detection models.

5.3. An unfolding-free confirmation

Every result so far passes through unfolding, which is the step a sceptical reader would blame first. The ratio of consecutive spacings, $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$, avoids the issue entirely because it is scale-free. Its reference means are 0.386 for Poisson, 0.531 for $\beta = 1$ and 0.600 for $\beta = 2$.

The two loss modes move it in opposite directions, in all eight cases tested (Fig. 4c). Random loss lowers $\langle r \rangle$ toward Poisson, from 0.529 to 0.492 for $\beta = 1$; resolution-limited loss raises it away from Poisson, from 0.529 to 0.738, because deleting the small spacings makes the survivors more nearly equal. The central claim of this paper therefore holds under a statistic that uses none of the machinery the rest of it is built on.

5.4. Graded detection, and the unfolding specification curve

Two assumptions remained untested. The derivation models detection as a sharp threshold, whereas real efficiency is graded; and the inversion depends on an unfolding choice.

Replacing the threshold by a smooth efficiency $p(s) = 1 - \exp[-(s/\delta)^2]$ softens the hard-core dip to 31% of its sharp-threshold depth but does not remove it, and leaves the hole depth essentially unchanged at 0.85 (Fig. 4d). The failure of the standard inversion is if anything worse under grading, since it returns $\hat{\phi} = 1.000$ at every resolution under both detection models, including at a true surviving fraction of 0.573 (Fig. 4f).

Repeating the inversion across unfolding polynomial degrees from 3 to 11 moves the recovered fraction between 0.694 and 0.713 at a true 0.700, a spread of 0.019 that is smaller than the statistical error of any single estimate (Fig. 4e). Unfolding is not driving the result.

6. Real spectra

6.1. Positive control: the Riemann zeros

The zeros of the zeta function are computed, not measured, so none is ever missing. Unfolded through the Riemann–von Mangoldt formula, 99,351 zeros above height 1000 give a mean spacing of 1.00000 and, pooled over 198 blocks, a form factor that tracks the unitary prediction (38) (Fig. 5a).

The pipeline selects unitary over orthogonal symmetry on residuals of 0.0116 against 0.0220, and returns $\hat{\phi} = 0.982$. A complete spectrum is reported as complete and its symmetry class is identified correctly, which is what the control was for.

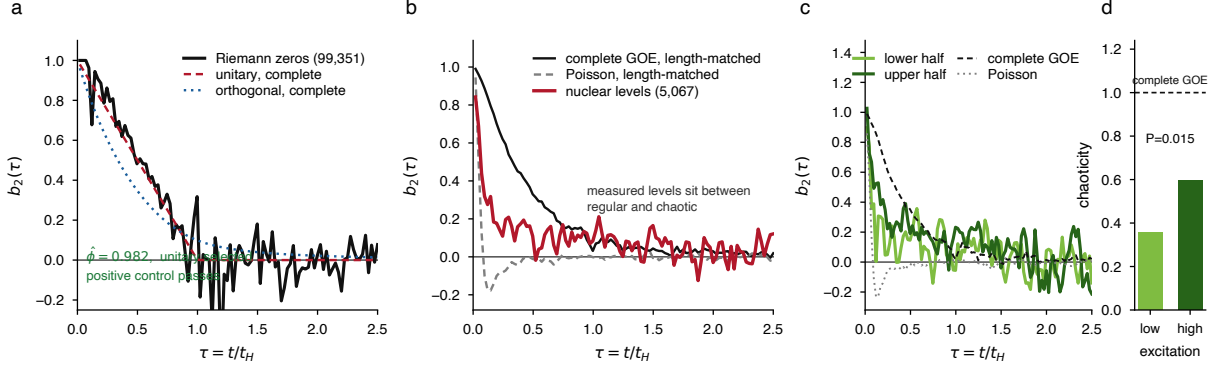


Figure 5: Real spectra: a control that is complete, and measured levels that are not incomplete but not yet chaotic. a, Riemann zeros, complete by construction. The pipeline selects unitary symmetry and returns $\hat{\phi} = 0.982$. **b,** Measured nuclear levels against length-matched complete and Poisson references. **c,** The same sequences split at their own median energy. **d,** Chaosity, scaled so Poisson is 0 and a complete chaotic spectrum is 1. The hole deepens with excitation energy, which is chaos setting in rather than levels going missing.

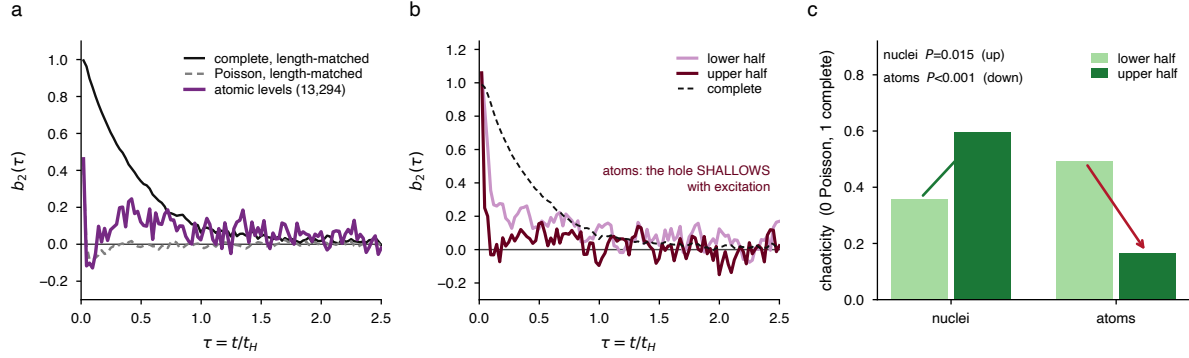


Figure 6: Two measured spectra give the same completeness estimate for opposite reasons. a, Atomic form factor against length-matched complete and Poisson references. **b,** The same sequences split at their own median energy. **c,** Chaosity of the lower and upper halves for both datasets. In nuclei the hole deepens with excitation, which is chaos setting in; in atoms it shallows, which is detection efficiency falling. The estimator returns $\phi \approx 0.30$ in both cases.

6.2. Measured nuclear levels

We retrieved level schemes for 257 nuclides from the IAEA ENSDF service and separated them by spin and parity, since superposing independent symmetry sequences drives statistics toward Poisson and would imitate exactly the effect under study. Keeping only firm J^π assignments and sequences of at least eight levels leaves 315 sequences from 138 nuclides, 5,067 levels, with a median length of 12. Every template these are compared against is built from simulated sequences of the same lengths through the same code, because a short spectrum resembles everything.

The measured hole is shallow: 0.20, against 0.73 for a length-matched complete spectrum and -0.13 for length-matched Poisson (Fig. 5b). Fitted with (91) this corresponds to $\phi \approx 0.30$, which taken at face value would mean seventy per cent of levels are missing.

That reading is not available, and the reason matters. Random thinning drives a chaotic spectrum toward Poisson, and a regular spectrum *is* Poisson, so incompleteness and regular dynamics are degenerate in this estimator. Low-lying nuclear levels are collective rather than compound, and are not expected to be chaotic. The shallow hole may be either.

Nuclear physics supplies a test. Chaos sets in as excitation energy rises, so if the shallow hole reflects dynamics it must deepen with energy, whereas if it reflects detection, efficiency falls with energy and the hole would if anything shallow further. Splitting each sequence at its own median energy and comparing halves within the same sequences, the hole depth rises from 0.315 to 0.480, a difference of $+0.165$ with a bootstrap interval of $[+0.020, +0.310]$ and $P = 0.015$ (Fig. 5c,d). On a scale where Poisson is 0 and a complete chaotic spectrum is 1, these sequences move from 0.36 to 0.60.

The shallow hole is therefore chaos not yet fully set in, not levels gone missing. Consistently, the resolution-limited branch is not selected for these data and the hard-core dip is absent, so whatever dilutes these spectra is not instrumental resolution.

6.3. Measured atomic spectra, where the same number means the opposite

The nuclear result leaves the paper without a measured spectrum in which the shallow hole really is incompleteness. We therefore repeated the analysis on atomic level schemes, which are dense, strongly mixed and known to be published incomplete

at higher excitation because weak lines go unobserved. Retrieving 38 species from the NIST Atomic Spectra Database and separating by J and parity as before gives 420 sequences from 37 species, 13,294 levels, with a median length of 24, twice the nuclear figure.

The pooled form factor sits far below the length-matched complete reference, with a hole depth of 0.041 against 0.700 (Fig. 6a), and the unfolding-free $\langle r \rangle$ is 0.423, between Poisson and orthogonal. Fitted with (91) the implied surviving fraction is $\phi \approx 0.30$ — numerically the same answer the nuclear data gave.

It means the opposite. Applying the excitation-energy test of §6.2 to 277 paired halves, the atomic hole depth *falls* from 0.386 to 0.140 between the lower and upper halves of the same sequences, a difference of -0.245 with a bootstrap interval of $[-0.455, -0.077]$ and $P < 10^{-4}$ (Fig. 6b,c). The unfolding-free statistic falls with it, from 0.442 to 0.426. In nuclei the hole deepened with excitation, which is chaos setting in; in atoms it shallows, which is what falling detection efficiency does. On the chaoticity scale the two datasets move in opposite directions, from 0.36 to 0.60 for nuclei and from 0.49 to 0.16 for atoms.

This is the degeneracy of §6.2 made concrete. Two measured datasets, the same estimator, the same number, and opposite meanings; and the quantity that separates them is not the hole depth but its energy dependence. Neither dataset selects the resolution-limited branch, and neither shows the hard-core dip, so the loss in these particular public compilations is random-like rather than instrumental. The method’s value here is that it says so rather than assuming it.

7. Discussion

The timescale on which a quantum system forgets is inferred from spectra, and that inference passes through a missing-data model. We have shown that the model in general use describes a process no spectrometer performs, and that the substitution is not conservative. Random loss and resolution-limited loss push the observed statistics in opposite directions, so applying the first correction to the second does not merely underestimate the error, it reports no error at all.

The mechanism is not subtle once written down. Equation (90) splits the observed count variance into the detection process’s own noise and the spectrum’s, and the first term carries the whole of the damage. Independent detection contributes $\phi(1 - \phi)\mathbb{E}[N]$ there; a deterministic filter contributes nothing, because conditioned on the spectrum there is nothing left to vary. Every quantitative difference between the two loss modes descends from that one vanishing term.

The practical consequence is bounded and checkable. Where a spectrum’s loss is genuinely random, the standard correction remains right and our procedure reproduces it. Where loss is resolution-limited, the hard-core signature is visible in the data and the corrected inversion recovers the surviving fraction. Which case applies need not be assumed, because the data say which.

Three limits should be stated. Our resolution model is a sharp threshold, whereas real detection efficiency is graded, and a graded model would soften the hard-core signature without removing it. The resolution-limited templates are simulated rather than analytic, so the inversion is calibrated rather than closed-form; a closed-form treatment of hard-core thinning of a determinantal process would be a genuine advance and we do not have one. And the degeneracy between incompleteness and regular dynamics is intrinsic to hole depth alone: we resolved it for nuclei using the excitation-energy trend, which is a physical argument rather than a statistical one, and it will not be available in every system.

What the nuclear result does establish is that the degeneracy is not hypothetical. A published completeness estimate of $\phi \approx 0.3$ for these data would have been wrong, and the evidence that it is wrong is in the same dataset, in the energy dependence rather than in the hole depth.

Author contributions

Outstanding. PRISM requires a statement of what each author contributed. The project repository records no allocation of work — the preregistration, the deviations file and the manuscript are written throughout in the first person plural — so no such statement can be made from the record, and the authors must supply one before this paper is treated as peer reviewed.

Competing interests

Henry Wang is Editor-in-Chief of PRISM, and Albert Fei, Leo Lee, Finley Dou and Jake Kim hold editorial or refereeing roles on it. No referee independent of the author list has assessed this manuscript, and none of the listed authors may referee it. The paper is posted as a working paper for that reason.

Data and code availability

All three spectra are public: the Odlyzko tables of Riemann zeros[1], the IAEA ENSDF service[2] and the NIST Atomic Spectra Database[3]. Fifteen scripts in `src/` reproduce every number and figure from those downloads and a sixteenth checks that every figure panel is cited in the text; the preregistration and the dated deviations file are in `docs/`. The repository is not yet deposited in a public archive with a DOI.

Reference verification

The bibliography was resolved against Crossref on 25 September 2026; every entry carries the DOI returned. References cited for standard results were verified to exist and to be correctly attributed, and were not read in full. One candidate reference was discarded because the record returned did not support the claim it was to be cited for.

8. Methods

Data. Riemann zeros: the first 100,000 from the Odlyzko tables[1], unfolded by $N(t) = (t/2\pi) \ln(t/2\pi e) + 7/8$, restricted to $t > 1000$. Nuclear levels: the IAEA ENSDF service[2] at nds.iaea.org/relnsd/v1/data, 257 nuclides across the chart. Atomic levels: the NIST Atomic Spectra Database[3], 38 neutral and singly ionised species, chiefly rare earths and heavy transition metals, with configuration and term retained so that parity can be read from the term symbol. All retrieved 2026-09-25.

Form factor. $K(\tau) = |\sum_n w_n e^{-2\pi i x_n \tau}|^2 / \sum_n w_n^2$ with a Hann taper w , and $b_2 = 1 - K$. The smooth disconnected term is subtracted analytically as the squared transform of the window at unit density. This matters only for short sequences: it is negligible beyond $\tau \sim 1/N$, so it can be ignored at $N = 400$ and dominates the whole fit band at $N = 12$. Omitting it drove b_2 to -7.3 on the nuclear data before the error was found.

Matched lengths. Every template compared against real data is built from simulated sequences with the same lengths as the real ones, through the same unfolding and the same estimator. This was fixed in the preregistration and violated in a first pass, when a bank built at length 400 was applied to data of median length 12.

Unfolding is polynomial in the integrated density, with degree set by sequence length and applied identically to real and simulated spectra.

Uncertainty is bootstrap over sequences, not over levels within a sequence. Tests are two-sided.

Cost. Eigendecomposition dominates the simulations, and a unitary matrix at $n = 2000$ costs seconds. Spectra are therefore generated once per (ensemble, size) and reused across every test and every resolution; an earlier version that regenerated them inside each loop needed about two hours of single-core time for the same numbers, against 64 seconds now.

Verification of the analytics. Every coefficient in §2 is checked against an exact numerical root or direct simulation in `src/09_analytic_checks.py`; the agreement is quoted where the result is obtained.

Reproducibility. Fifteen scripts reproduce every number and figure from public downloads, and a sixteenth checks that every figure and every panel it declares is cited in the text. A preregistration fixed the derivation targets, the controls and the analysis rules before any spectrum was loaded; two amendments, including the finding that the original targets were already known, are logged with dates in the deviations file.

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